Binyan Xu

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Education

The Chinese University of Hong Kong — **Sep. 2023 – Aug. 2027**
Ph.D. student in Information Engineering — GPA: 3.86

University of California, Berkeley — **Aug. 2021 – Dec. 2021**
Exchange Program in EECS — GPA: 4.0

Xi'an Jiaotong University — **Sep. 2019 – Jul. 2023**
B.Sc. in Automation — Honor, GPA: 3.92

First Author Publications: 4 CCF-A

Binyan Xu *et al.*. From Multi-Agent to Single-Agent: When Is Skill Distillation Beneficial? Under Review.

Binyan Xu *et al.*. Contextual Agentic Memory is a Memo, Not True Memory. Under Review.

Binyan Xu *et al.*. Reviewer Scores Are Not Comparable Across Research Areas in ML Peer Review. Under Review.

Binyan Xu *et al.*. From Internal Diagnosis to External Auditing: A VLM-Driven Paradigm for Data-Free Online Backdoor Defense. In *ICML '26* **(CCF-A)**.

Binyan Xu *et al.*. Breaking the Stealth-Potency Trade-off in Clean-Image Backdoors with Generative Trigger Optimization. In *AAAI '26*, 2026, **Oral Presentation (CCF-A)**.

Binyan Xu *et al.*. One Surrogate to Fool Them All: Universal, Transferable, and Targeted Adversarial Attacks with CLIP. In *CCS '25*, 2025, **Oral Presentation (CCF-A)**.

Binyan Xu *et al.*. CLIP-Guided Backdoor Defense through Entropy-Based Poisoned Dataset Separation. In *MM '25*, 2025, **Oral Presentation (CCF-A)**.

Internships

Agentic AI Research on Single-Agent and Multi-Agent Systems — **Jan. 2026 – Present**
Tencent, *Qingyun Program Internship* | LLM Agent, Skill Distillation, Multi-Agent System
- Studied when distilling a multi-agent system (MAS) into a single-agent skill yields performance gains.
- Proposed Metric Freedom, a principled predictor of skill utility measured before any skill is built.
- Adaptive skill matches or exceeds original MAS at 1.4–15× lower cost across 4 tasks and 11 datasets.
- 1 paper submitted to NeurIPS (Under Review).

AI-Aided Ascetic Graphic Design Generation — **Jul. 2022 – Jun. 2023**
Microsoft Research Asia (MSRA), *Stars of Tomorrow* | Diffusion Model, Large Language Model
- Developed an ascetic graphic design system with an autoregressive vision-language model for layout generation.
- Enhanced this graphic design system with diffusion model to generate additional decorative patterns.
- Significantly outperforms previous approaches in both better generation quality and less constraint violation.

Projects

Foundation Model-Aided Neural Network Attacks and Defenses — **Sep. 2023 – Present**
Laboratory for Applied Security Research, CUHK | Backdoor, Adversarial Attack, VLM, Diffusion Model
- Proposed an InfoGAN-based stealthy backdoor attack, achieving $\geq 90\%$ ASR with $\leq 1\%$ clean accuracy drop.
- Involved in UnivIntruder, a transferable attack achieving 99.4% ASR across 4 datasets and 84% on Google/Baidu.
- Proposed CLIP-guided backdoor defense via entropy-based data separation, defending all kinds of backdoor attacks.
- Proposed a VLM-driven data-free online backdoor defense paradigm leveraging VLMs as external auditors.
- 3 papers accepted (CCS '25, MM '25, AAAI '26, all CCF-A Oral); 1 paper submitted to ICML '26 (avg. score 4.25).

Selected Awards

- Finalist Award, Mathematical Contest in Modeling
- Qian Xuesen Program Honorary Graduate
- Outstanding Graduates Scholarship
- First-class Funding for Studying Abroad (about ¥100k)

Skills

Languages: Chinese (Native), English (Fluent); TOEFL: 104 (R29/L28/S22/W25), CET-6: 584
Programming: Python, PyTorch, TensorFlow, C/C++, MATLAB